250 Homework #2 (Fall 2026)

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This assignment is due at 11:59 p.m. on Friday 9 October 2026, to be submitted on Gradescope. Late assignments (with penalty, unless excused) will be accepted until 11:59 p.m. on Saturday 10 October 2026, and not after. Please see the syllabus for more details on homework grading and late penalties.

When submitting to Gradescope, please do not forget to assign pages to problems.

All problems, with the exception of the first two and , originally come from the textbook. Some of them have been modified. In case of doubt, the version printed here is the one you should follow.

The instructions for this homework are the same as for HW1. Please read them there.

See the Canvas Homework web page for detailed instructions on how to format solutions. Solutions that do not follow those formatting instructions will be returned for rewriting and resubmission. In later assignments there will be points deducted for such formatting errors.

*Collaborated with Nobody.

Question 1 [10 pts]

Your solution should follow the format given on pages 17-19 of the relation study guide.

Let $A = \{a, b, c, d, e\}$. The relation $P \subseteq A \times A$ is a **partial order**.
It is known that

1. $(a, b), (b, c), (d, e), (d, b) \in P$ and
2. $(c, e), (e, b), (d, a), (b, d) \notin P$.

(i) In the table given below, label all pairs that must be in P with a "✓" and all pairs that cannot be in P with a "×." For pairs that are neither, leave their entry blank. We start you off by labelling the five known pairs with the appropriate "✓" or ×.

For full credit, you must justify your answers. Do that by numbering your justifications and, next to each ✓ or × in the table, write the number of the justification for that entry.

(a, a)		(b, a)		(c, a)		(d, a)	×2		(e, a)		
(a, b)	✓1	(b, b)		(c, b)		(d, b)	✓1		(e, b)	×2	
(a, c)		(b, c)	✓1	(c, c)		(d, c)			(e, c)		
(a, d)		(b, d)	×2	(c, d)		(d, d)			(e, d)		
(a, e)		(b, e)		(c, e)	×2	(d, e)	✓1		(e, e)		

(ii) How many partial orders P' exist such that (a, b), (b, c), (d, e) and (d, b) are in P', and that satisfy that (c, e), (e, b), (d, a), and (b, d) are not in P'? Briefly explain how you know this.

Draw a separate Hasse diagram for each such partial order P'.
Note. You must draw these using the convention taught in class: an edge corresponding to relation (a, b) must go *up* from a to b.

Solution:

(i)

(a, a)	✓3	(b, a)	×5	(c, a)	×5	(d, a)	×2		(e, a)	×6	
(a, b)	✓1	(b, b)	✓3	(c, b)	×5	(d, b)	✓1		(e, b)	×2	
(a, c)	✓4	(b, c)	✓1	(c, c)	✓3	(d, c)	✓4		(e, c)		
(a, d)		(b, d)	×2	(c, d)	×5	(d, d)	✓3		(e, d)	×5	
(a, e)		(b, e)		(c, e)	×2	(d, e)	✓1		(e, e)	✓3	

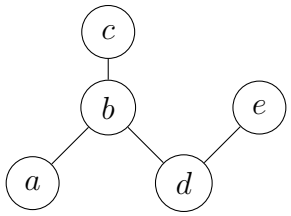
1. Its given that $(a, b), (b, c), (d, e), (d, b) \in P$
2. Its given that $(c, e), (e, b), (d, a), (b, d) \notin P$.
3. Since its a partial order, it must be reflexive; therefore every identity pair must be present: $\forall x \in A, (x, x) \in P$; $(a, a), (b, b), (c, c), (d, d), (e, e) \in P$

4. Since its a partial order, it must be transitive; therefore for any pairs in (1.) where the output is the same as another pair's input, the first pair's input must be paired with the second pair's output:
   a. Since $(a, b), (b, c) \in P$; $(a, c) \in P$
   b. Since $(d, b), (b, c) \in P$; $(d, c) \in P$
5. Since its a partial order, it must be antisymmetric; therefore every known pair from (1.) and (4.) CANNOT have a matching reverse pair (unless the elements are identical like identity pairs): $(b, a), (c, b), (e, d), (b, d), (c, a), (c, d) \notin P$
6. From transitivity, any pairs that would require a pair we've already ruled out to be present, cannot be in P; Since $(a, b) \in P$, and $(e, b) \notin P$; $(e, a) \notin P$

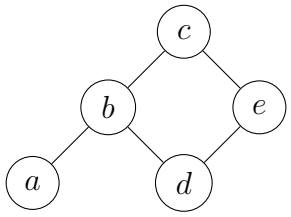
The entries for $(a, d), (a, e), (b, e), (e, c)$ are left blank because the given facts cannot be used to definitively derive whether those pairs *must* or *cannot* be in P.

(ii) There are 16 theoretical combinations of the four optional pairs (2^4), but only **10** of those 16 are actually possible with the given conditions.

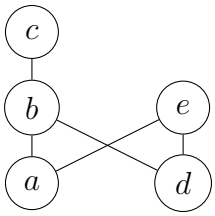
P01: $(a, d), (a, e), (b, e), (e, c) \notin P$
This is the minimal partial order with none of the four optional pairs included. Transitivity provides the (a, c) and (d, c) pairs in the graph.



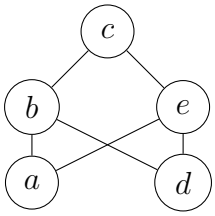
P02: $(a, d), (a, e), (b, e) \notin P$ and $(e, c) \in P$
This partial order includes 1 of the four optional pairs (e, c). None of the other optional pairs need to be included through transitivity. Transitivity provides the $(a, c), (d, c)$ pairs in the graph.



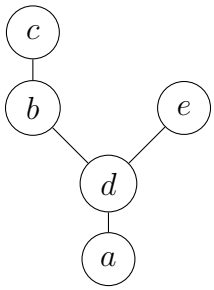
P03: $(a, d), (e, c), (b, e) \notin P$ and $(a, e) \in P$
This partial order includes 1 of the four optional pairs (a, e). None of the other optional pairs need to be included through transitivity. Transitivity provides the $(a, c), (d, c)$ pairs in the graph.



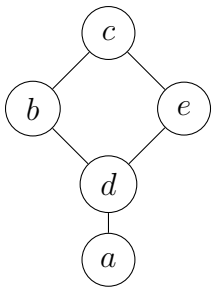
P04: $(a, d), (b, e) \notin P$ and $(a, e), (e, c) \in P$
This partial order includes the 2 optional pairs $(a, e), (e, c)$ that can be flipped independently of the other optional pairs (as seen in P03 and P02). Transitivity provides the $(a, c), (d, c)$ pairs in the graph.



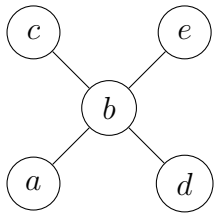
P05: $(b, e), (e, c) \notin P$ and $(a, d), (a, e) \in P$
Including (a, d) REQUIRES including (a, e) through transitivity, since (d, e) exists in all variations of P. Transitivity provides the $(a, c), (a, b), (a, e), (d, c)$ pairs in the graph.



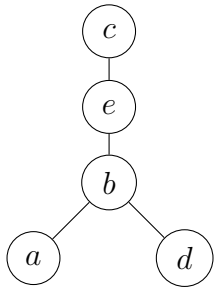
P06: $(b, e) \notin P$ and $(a, d), (a, e), (e, c) \in P$
This is the same as P04, just with the independent (e, c) included as well. Transitivity provides the $(a, c), (a, b), (a, e), (d, c)$ pairs in the graph.



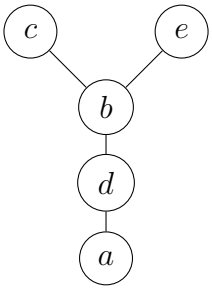
P07: $(a,d),(e,c) \notin P$ and $(b,e),(a,e) \in P$
Including (b,e) REQUIRES including (a,e) through transitivity, since (a,b) exists in all variations of P. Transitivity provides the $(a,c),(d,e),(a,e),(d,c)$ pairs in the graph.



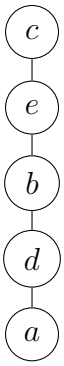
P08: $(a,d) \notin P$ and $(b,e),(a,e),(e,c) \in P$
This is the same as P06, just with the independent (e,c) included as well. Transitivity provides the $(a,c),(d,e),(a,e),(d,c),(b,c)$ pairs in the graph.



P09: $(e,c) \notin P$ and $(a,d),(b,e),(a,e) \in P$
This is the same as P06, but since (a,d) ALSO requires including (a,e), we can make a version with all 3. Transitivity provides the $(a,c),(a,b),(a,e),(d,c),(d,e)$ pairs in the graph.



P10: $(e,c),(a,d),(b,e),(a,e) \in P$
This is the same as P08, just with the independent (e,c) included as well. Transitivity provides the $(a,c),(a,b),(a,e),(d,c),(d,e),(d,c)$ pairs in the graph.


c
e
b
d
a

Question 2 [10 pts]

Your solution should follow the format given on pages 20-23 of the relation study guide.

Let $A = \{a, b, c, d, e, f, g\}$. The relation $Q \subseteq A \times A$ is an **equivalence relation**. It is known that

1. $(a, b), (c, d), (e, f) \in Q$, and
2. $(c, f), (g, e) \notin Q$.

(i) In the table given below, label all pairs that must be in Q with a "✓" and all pairs that cannot be in Q with a "×." For pairs that are neither, leave their entry blank. We start you off by labelling the five known pairs with the appropriate "✓" or ×.

For full credit, you must justify your answers. Do that by numbering your justifications and, next to each ✓ or × in the table, write the number of the justification for that entry.

(a,a)			(b,a)			(c,a)			(d,a)			(e,a)			(f,a)			(g,a)	
(a,b)	✓1		(b,b)			(c,b)			(d,b)			(e,b)			(f,b)			(g,b)	
(a,c)			(b,c)			(c,c)			(d,c)			(e,c)			(f,c)			(g,c)	
(a,d)			(b,d)			(c,d)	✓1		(d,d)			(e,d)			(f,d)			(g,d)	
(a,e)			(b,e)			(c,e)			(d,e)			(e,e)			(f,e)			(g,e)	×2
(a,f)			(b,f)			(c,f)	×2		(d,f)			(e,f)	✓1		(f,f)			(g,f)	
(a,g)			(b,g)			(c,g)			(d,g)			(e,g)			(f,g)			(g,g)	

(ii) (a) How many equivalence relations Q' exist that satisfy $(a, b), (c, d), (e, f) \in Q'$, and $(c, f), (g, e) \notin Q'$?

(b) Also, write down, in partition format, all of the equivalence relations that satisfy the 5 known pairs. It is not necessary to show how you derived them.

When writing down a partition, each of the subsets should be written in alphabetical order with each set being written in alphabetical order and the order of the sets being the alphabetical ordering of their 'first' element. For example, the equivalence relation that has g, c, e together, d, b together and a, f together should be written as $\Big\{\{a, f\},\ \{b, d\},\ \{c, e, g\}\Big\}$.

Solution:

(i)

(a,a)	✓3		(b,a)	✓5		(c,a)			(d,a)			(e,a)			(f,a)			(g,a)	
(a,b)	✓1		(b,b)	✓3		(c,b)			(d,b)			(e,b)			(f,b)			(g,b)	
(a,c)			(b,c)			(c,c)	✓3		(d,c)	✓5		(e,c)	×5		(f,c)	×5		(g,c)	
(a,d)			(b,d)			(c,d)	✓1		(d,d)	✓3		(e,d)	×5		(f,d)	×5		(g,d)	
(a,e)			(b,e)			(c,e)	×4		(d,e)	×4		(e,e)	✓3		(f,e)	✓5		(g,e)	×2
(a,f)			(b,f)			(c,f)	×2		(d,f)	×4		(e,f)	✓1		(f,f)	✓3		(g,f)	×6
(a,g)			(b,g)			(c,g)			(d,g)			(e,g)	×5		(f,g)	×6		(g,g)	✓3

1. Its given that $(a,b),(c,d),(e,f) \in Q$
2. Its given that $(c,f),(g,e) \notin Q$
3. Since its an equivalence relation, it must be reflexive; therefore every identity pair must be present: $\forall x \in A, (x,x) \in Q$; $(a,a),(b,b),(c,c),(d,d),(e,e),(f,f),(g,g) \in Q$
4. Since its an equivalence relation, it must be transitive; therefore any pairs that would require a pair we've already excluded to be included, cannot be in Q since they imply a pair that CANNOT be included:
   - Since $(c,d) \in Q$, and $(c,f) \notin Q$; $(d,f) \notin Q$
   - Since $(e,f) \in Q$, and $(c,f) \notin Q$; $(c,e) \notin Q$
   - Since $(c,d),(e,f) \in Q$, and $(c,f) \notin Q$; $(d,e) \notin Q$
5. Since its an equivalence relation, it must be symmetric; therefore every known pair from (1.) MUST have a matching reverse pair included, and every known pair from (2.) and (4.) CANNOT have a matching reverse pairs included: $(b,a),(d,c),(f,e) \in Q$, and $(f,c),(e,g),(f,d),(e,c),(e,d) \notin Q$
6. Just like in (4.), because of transitivity, since $(e,f) \in Q$, and $(e,g) \notin Q$; $(f,g) \notin Q$; Also, just like in (5.), because of symmetry, $(g,f) \notin Q$

The remaining entries are left blank because the given facts cannot be used to definitively derive whether those pairs *must* or *cannot* be in Q.

(ii) (a) From part (i), the initial equivalence classes are $\{a,b\},\{c,d\},\{e,f\},\{g\}$.
Since (c,f) is excluded, the classes $\{c,d\}$ and $\{e,f\}$ cannot be merged. Similarly, since (g,e) is excluded, the classes $\{e,f\}$ and $\{g\}$ cannot be merged. With the given conditions, exactly **7** possible combinations exist.

(b)
1. $\Big\{\{a,b\},\{c,d\},\{e,f\},\{g\}\Big\}$ (The initial relation)
2. $\Big\{\{a,b,c,d\},\{e,f\},\{g\}\Big\}$ (Merge a,b and c,d)
3. $\Big\{\{a,b,e,f\},\{c,d\},\{g\}\Big\}$ (Merge a,b and e,f)
4. $\Big\{\{a,b,g\},\{c,d\},\{e,f\}\Big\}$ (Merge a,b and g)
5. $\Big\{\{a,b\},\{c,d,g\},\{e,f\}\Big\}$ (Merge c,d and g)
6. $\Big\{\{a,b,c,d,g\},\{e,f\}\Big\}$ (Merge a,b and c,d and g)
7. $\Big\{\{a,b,e,f\},\{c,d,g\}\Big\}$ (Merge a,b and e,f; Merge c,d and g)

Question 3. P2.1.7 [10 pts]

Let X be a set of at least two people. We will ask each person in X to write down the name of a person in X other than themself. Let R be the relation $\{\langle a, b\rangle : a\text{'s chosen name is } b\}$. Suppose we start with a given person a, move to the person that a wrote down, then to the person that they wrote down, and so forth until we reach a given person b. Let S be the relation

$$S = \{\langle a, b\rangle : \text{Starting from } a, \text{ we eventually reach } b \text{ by this process}\}.$$

(a) If from a, we reach a again before finding b, explain why we will never reach b.

(b) If $\langle a, b\rangle$ is not in S, when can we give up on the process, i.e., be sure that $\langle a, b\rangle$ is not in S.

(c) If $\langle a, b\rangle$ is not in S, will we definitely reach a. i.e., is $\langle a, a\rangle$ in S? If the answer is yes, prove it. If the answer is no, give a counterexample in which $\langle a, b\rangle \notin S$ but $\langle a, a\rangle \notin S$.

Solution:

(a) If from a, we reach a again before finding b, we will never reach b. This is because if we circle back to a, it must be a cycle/loop since each person only writes down a single name (each input only has 1 output); and since a loop repeats infinitely, we would never get the chance to reach b, which lies OUTSIDE the loop.

(b) We can give up on the process and conclude that $\langle a, b\rangle \notin S$ if and ONLY if, we see a person's name more than once before finding b. This is because seeing a person's name more than once signifies an infinite loop, and if we hadn't found b by then, it'll never come up in the infinite loop that is made up of people we've already checked.

(c) If $\langle a, b\rangle$ is not in S, that does NOT necessarily mean that $\langle a, a\rangle$ is in S. Just because an infinite loop is reached, does not mean that it has to include all the way back as far as a; for example, if a wrote down x, x write down y, and y wrote down x, then even though you started with a and got caught in an infinite loop, the loop is just going back and forth between x and y.

Question 4. P2.3.3 [10 pts]

Let A and B be two types such that A is a proper subset of B ($A \subset B$, i.e., so that B contains all the elements of A plus at least one other element). Let P be a unary predicate on A, and let Q be a unary predicate on B. Suppose that $\forall a : \forall b : P(a) \rightarrow Q(b)$ is true.

Which of the following four statements is guaranteed to be true? For each statement, explain why it is always true or give an example where it is false. (Hint: Consider the case where A is empty.)

(a) $(\forall a : P(a)) \rightarrow (\forall b : Q(b))$

(b) $(\forall b : Q(b)) \rightarrow (\forall a : P(a))$

(c) $(\exists a : P(a)) \rightarrow (\exists b : Q(b))$

(d) $(\exists b : Q(b)) \rightarrow (\exists a : P(a))$

Solution:

In English, the given condition that is always true is: For every element of A and B, if the element of A satisfies P, then the element of B must satisfy Q

(a) FALSE
In English, if every element of A satisfies P, then every element of B satisfies Q.
Although it sounds nearly identical to the given condition, consider the empty set for A; P will always evaluate to TRUE since there are no counterexamples, meanwhile if even 1 value of B (which must have at least 1 value in a addition to the empty set) doesn't satisfy Q, then the implication is false. For example: P(x) is defined as "*x is not equal to 67*", Q(x) is defined as "*x is equal to 2*", A is the empty set, B is the set 1;

(b) FALSE
In English, if every element of B satisfies Q, then every element of A satisfies P.
According to the given condition, whether or not elements of B satisfy Q has no bearing on whether or not elements of A satisfy P. For example: P(x) is defined as "*x is less than 1*", Q(x) is defined as *x is greater than 5*, A is the set 6, B is the set 6,7.

(c) TRUE
In English, if at least one element of A satisfies P, then at least one element of B satisfies Q.
Since there needs to be at least one element in A for this proposition, the empty set trick won't work this time, if we do set A to the empty set it will just be false since there isn't at least one element. With the given condition, it is impossible for an element of A to satisfy P without an element of B satisfying Q, therefore this statement is guaranteed to be true.

(d) FALSE

In English, if at least one element of B satisfies Q, then at least one element of A satisfies P.

According to the given condition, whether or not elements of B satisfy Q has no bearing on whether or not elements of A satisfy P. For example: P(x) is defined as "*x is less than 1*", Q(x) is defined as *x is equal to 7*, A is the set 6, B is the set 6,7.

Question 5. Modified from P2.3.8 [10 pts]

Let X be a set and $R(x)$ a unary predicate on X.

(a) Write a quantified statement that says "there is exactly one element x in X such that $R(x)$ is true".

(b) Write a quantified statement that says "there are exactly two elements x in X such that $R(x)$ is true".

(c) Write a quantified statement that says "there are at least three elements x in X with $R(x)$ true, and at least three with $R(x)$ false"

Note: (a) is so common that there is shorthand macro for writing it. " $\exists! x : R(x)$" denotes that there is a unique $x \in X$ satisfying $R(x)$, which is what (a) is stating. This problem is asking you how to write this from scratch.

Your statements may use the binary relations $x = y$ and $x \neq y$).

Solution:

(a) $\exists a \in X : (R(a) \wedge (\forall c \in X : (c \neq a) \rightarrow \neg R(c)))$

(b) $\exists a, b \in X : (R(a) \wedge R(b) \wedge (a \neq b) \wedge (\forall c \in X : (c \neq a \wedge c \neq b) \rightarrow \neg R(c)))$

(c) $\exists a, b, c, d, e, f \in X : (R(a) \wedge R(b) \wedge R(c) \wedge \neg R(d) \wedge \neg R(e) \wedge \neg R(f) \wedge (a \neq b) \wedge (a \neq c) \wedge (b \neq c) \wedge (d \neq e) \wedge (d \neq f) \wedge (e \neq f))$

Question 6. P2.6.10 [10 pts]

One day, a set of five dogs met after their morning walks to compare their observations of birds. The set D of dogs consisted exactly of Arly (a), Cardie (c), Duncan (d), Mia (m), and Whistle (w).

The set S of bird species in question consisted exactly of Bluebird (B), Crow (C), Heron (H), Mallard (M), and Woodpecker (W).

The predicate $R(x, y)$ means "dog x observed a bird of species y".

(a) Translate the following statements into symbolic language.

I. There was a species that was observed by all the dogs.

II. Arly observed a Crow and no other species.

III. Every dog other than Arly observed at least two different species.

IV. Each dog, other than Arly and Cardie, observed all the species that Cardie observed and at least one that Cardie did not. (Note: Different dogs may have observed different species that Cardie did not.)

V. Duncan was the one and only dog to observe a Woodpecker, and Whistle was the one and only dog to observe a Bluebird.

(b) Use statements (I)-(V) to prove that some dog observed a Heron, i.e., $\exists x : R(x, H)$.

Your proof should provide all steps. In particular, it should explicitly state when Instantiation, Specialization, etc are used.

Solution:

(a) I. $\exists y \in S : \forall x \in D : R(x, y)$

II. $R(a, C) \wedge \neg R(a, B) \wedge \neg R(a, H) \wedge \neg R(a, M) \wedge \neg R(a, W)$

III. $\forall x \in D : (x \neq a \rightarrow (\exists y, z \in S : (y \neq z) \wedge R(x, y) \wedge R(x, z)))$

IV. $\forall x \in D : (x \neq a \wedge x \neq c) \rightarrow (\exists y \in S : (R(x, y) \wedge \neg R(c, y)) \wedge \forall z \in S : (R(c, z) \rightarrow R(x, z)))$

V. $R(d, W) \wedge R(w, B) \wedge (\forall x \in D : (x \neq d \rightarrow \neg R(x, W)) \wedge (x \neq w \rightarrow \neg R(x, B)))$

(b) We want to prove that $\exists x \in D : R(x, H)$.

1. Statement (I) says theres a species that every dog observed. By instantiation, this species can be s. $\forall x \in D : R(x, s)$
2. Statement (II) says that Arly observed only a Crow, and since theres a species that EVERY dog observed, the species MUST be Crow, as its the ONLY one Arly observed. By substitution, $s = C$ and $\forall x \in D : R(x, C)$

3. Statement (V) says that Duncan was the only dog to observe Woodpecker, and Whistle was the only dog to observe Bluebird. Statement (III) says that every dog other than Arly observed at least two different Species. Cardie is not Arly, Duncan, or Whistle, so Cardie MUST have seen either Heron or Mallard (or both) in addition to Crow (which everyone has seen, given by step 2): $R(c, H) \vee R(c, M)$
4. (i) Case 1 from step 3 supposes $R(c, H)$; Since $c \in D$, generalization gives $\exists x \in D : R(x, H)$
   (ii) Case 2 from step 3 supposes both $R(c, H) \wedge R(c, M)$; Since $c \in D$, generalization gives $\exists x \in D : R(x, H)$
   (iii) Case 3 from step 3 supposes $R(c, M) \wedge \neg R(c, H)$. By statement (IV), every dog other than Arly and Cardie observed at least one species that Cardie did not (Heron, Bluebird, and Woodpecker). Duncan could have just observed a Woodpecker in addition, and Whistle could've just observed a Bluebird in addition; but that still leaves Mia, who couldn't have observed a Bluebird or Woodpecker, since Whistle and Duncan respectively were the ONLY dogs to observe them; therefore the only remaining species Mia could have observed in addition to the specias Cardie observed is Heron: thus $R(m, H)$ must be true, and since $m \in D$, generalization gives $\exists x \in D : R(x, H)$
5. Proof by cases proved that in all 3 cases, $\exists x \in D : R(x, H)$ is true; therefore $\exists x \in D : R(x, H)$

Question 7. P2.9.5 [10 pts]

Note: Be aware that, in the textbook, this problem spans two pages. Also, the solution to this one is long.

Let $A = \{v, w\}$ and $B = \{x, y, z\}$ be sets of characters.

(a) List all the possible functions from A to B. Determine which are injections.

(b) List all the possible functions from B to A. Determine which are surjections.

(c) How many possible functions are there from A to A? How many can be expressed as $g \circ f$, where f is a function from A to B and g is a function from B to A?

(d) How many possible functions are there from B to B? How many can be expressed as $f \circ g$, where g and f are functions as above?

Solution:

(a) A function is an injection when the distinct inputs have distinct outputs (when $f(v) \neq f(w)$).

1. $\{(v, x), (w, x)\}$ – NOT injection
2. $\{(v, x), (w, y)\}$ – injection
3. $\{(v, x), (w, z)\}$ – injection
4. $\{(v, y), (w, x)\}$ – injection
5. $\{(v, y), (w, y)\}$ – NOT injection
6. $\{(v, y), (w, z)\}$ – injection
7. $\{(v, z), (w, x)\}$ – injection
8. $\{(v, z), (w, y)\}$ – injection
9. $\{(v, z), (w, z)\}$ – NOT injection

(b) A function is a surjection if every element of the set the function is being applied *TO* is present in the outputs (both v and w appear as outputs).

1. $\{(x, v), (y, v), (z, v)\}$ – NOT surjection
2. $\{(x, v), (y, v), (z, w)\}$ – surjection
3. $\{(x, v), (y, w), (z, v)\}$ – surjection
4. $\{(x, v), (y, w), (z, w)\}$ – surjection
5. $\{(x, w), (y, v), (z, v)\}$ – surjection
6. $\{(x, w), (y, v), (z, w)\}$ – surjection

7. $\{(x, w), (y, w), (z, v)\}$ – surjection
8. $\{(x, w), (y, w), (z, w)\}$ – NOT surjection

(c) There are 4 possible functions from A to A (2^2), and all 4 can be expressed as $g \circ f$, where f is a function from A to B and g is a function from B to A; this is because f and g can both produce 2 outputs, exactly what ALL A to A functions need.

(d) There are 27 possible functions from B to B (3^3), but only 21 can be expressed as $f \circ g$, where f is a function from A to B and g is a function from B to A; this is because f and g can both only produce 2 outputs, so you have to exclude the 6 functions from B to B that have 3 distinct outputs.

Question 8. P3.1.4 [10 pts]

The **least common multiple** of two naturals x and y is the *smallest* natural that both x and y divide. For example, $lcm(8, 12) = 24$ because 8 and 12 each divide 24, and there is no smaller natural that both 8 and 12 divide.

(a) Find the least common multiple of 60 and 339.
Write it out as a number and not, e.g., as its prime factorization.

(b) Find the least common multiple of $2^3 \cdot 3^2 \cdot 5^4$ and $2^2 \cdot 3^4 \cdot 5^3$.
Write it out as a number and not, e.g., as its prime factorization

(c) Describe a general method to find the least common multiple of two naturals, given their factorization into primes (and assuming that the factorization exists and is unique). Write your solution in terms of $p_i(m)$ and $p_i(n)$ as described below.

Notation: Let p_i be the i'th prime. For natural m, let $p_i(m)$ be the exponent of p_i in the prime factorization of m, i.e., the largest power in p_i that divides m. So $m = \prod_{i=1}^{\infty} p_i^{p_i(m)}$. Example: $p_1 = 2$, $p_2 = 3$ and $p_3 = 5$. Since $20 = 2^2 \cdot 5^1$, $p_1(20) = 2$, $p_2(20) = 0$, $p_3(20) = 1$ and, $\forall i > 3$, $p_i(20) = 0$.

Solution:

(a) $60 = 2^2 \cdot 3 \cdot 5$ and $339 = 3 \cdot 113$
Therefore, $2^2 \cdot 3 \cdot 5 \cdot 113 = 6780$

(b) $2^3 \cdot 3^4 \cdot 5^4 = 405000$

(c) $lcm(m, n) = \prod_{i=1}^{\infty} p_i^{\max(p_i(m), p_i(n))}$
For each prime, choose the larger exponent, then multiply the prime powers together.

Extra Credit. P2.1.10 in the book. [10 pts]

Let C be a set of n candidates and J be a set of n jobs. Define the relation Q to be

$$Q = \{\langle c, j\rangle \ : \text{ candidate } c \text{ is qualified for job } j\}$$

Define the relation A to be

$$A = \{\langle c, j\rangle \ : \text{ candidate } c \text{ is assigned to job } j\}.$$

(a) We say that candidate c is "possible" for job j if they are qualified for j, they are not assigned to any other job, and no one else is assigned to j. Informally describe how you would determine, given boolean arrays for the relations Q and A, whether c is possible for j.
Note: This problem is essentially asking you to design an algorithm to verify *whether a given assignment A of people to jobs is possible.*

(b) One way to assign candidates to jobs would be to start with an empty assignment and repeatedly find a possible candidate for a job and assign them to it, until there are no possible candidates left. This **greedy algorithm** will not necessarily assign the greatest possible number of candidates to jobs. Find an example where there is a possible assignment of qualified candidates for all the jobs, but there is a sequence of assignments of possible candidates that stops, with no further assignments possible, while some jobs are still unfilled.
Note: This is asking you to design just one *counterexample. The counterexample could be very small, e.g., only 2 or 3 people and jobs.*

Solution:

(a) (1.) Check whether $Q(c, j)$ is true, meaning that candidate c is qualified for job j. If it is false, then c is not possible for j; Otherwise...

(2.) Check whether $A(c, k)$ is false where k is all jobs other than j, meaning that candidate c has no other jobs. If even one relation A is true and shows that candidate c is assigned another job, then c is not possible for j; Otherwise... candidate c is possible for j.

(b) Consider two candidates $c1, c2$ and two jobs $j1, j2$ with $Q = (c1, j1), (c1, j2), (c2, j1)$ ($c1$ is qualified for both jobs while $c2$ is only qualified for $j1$). Going in sequence order, $c1$ would get assigned $j1$ since they are qualified with no other assignment, $c1$ would NOT get assigned $j2$ since they already have another assignment, $c2$ would NOT get assigned $j1$ since it is already assigned, $c2$ would NOT get assigned $j2$ since they are not qualified. Thus, even though $c2$ could be assigned $j1$, and $c1$ could be assigned $j2$, the greedy algorithm did not see it happen.